

High-velocity resistance training improves executive function in mobility-limited older adults

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HIGHLIGHTS

- Short-term HVRT intervention improved the executive function of mobility-limited older adults.
- Changes in lower-body muscle parameters and functional capacity were associated with changes in executive function.
- Improvements in lower-body muscle parameters mediated the association between changes in executive function and functional performance.

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ABSTRACT

Objectives: To examine the effect of high-velocity resistance training (HVRT) on the executive function of middle-aged and older adults with and without mobility limitations.

Methods: Participants ($n = 41$, female: 48.9%) completed a supervised 12-week HVRT intervention (2 sessions/week; at 40–60% of one-repetition maximum). The sample included 17 middle-aged adults (40–55 years); 16 older adults (>60 years) and 8 mobility-limited older adults (LIM). Executive function was assessed before and after the intervention period and was reported as z-scores. Maximal dynamic strength, peak power, quadriceps muscle thickness, maximal isometric voluntary contraction (MVIC), and functional performance were also measured pre and post intervention. Training-related adaptations in cognitive measures were calculated using a Generalized Estimating Equation model.

Results: HVRT improved executive function in LIM (adjusted marginal mean differences [AMMD]: 0.21; 95%CI: 0.04, 0.38; $p = 0.040$) although no effect on middle-aged (AMMD: 0.04; 95%CI: -0.09; 0.17; $p = 0.533$) and older (AMMD: -0.11; 95%CI: -0.25; 0.02; $p = 0.107$) participants was observed. Improvements in maximal dynamic strength, peak power, MVIC, quadriceps muscle thickness, and functional performance were all associated with changes in executive function, and changes in the first four also seem to mediate the association between changes in functional performance and executive function.

Conclusions: HVRT-induced improvement in executive function of mobility-limited older adults were mediated by changes in lower-body muscle strength, power, and muscle thickness. Our findings reinforce the relevance of muscle-strengthening exercises to preserve cognition and mobility in older adults.

1. Introduction

Aging is marked by progressive impairments in neuromuscular,

physical, and cognitive function, all of which are partially secondary to deterioration of the neural system (Clark et al., 2013). This age-related decline can lead to limitations in older adults that hinder their capacity

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to perform common activities of daily living such as walking, rising from a chair, or climbing a flight of stairs. Such mobility impairments are typically associated with a greater risk of falling, developing cardiovascular disease, becoming disabled, and even dying (Newman et al., 2006). Because of populational aging, the prevalence of older adults with limited mobility is expected to rise unless interventions for preserving and improving mobility are identified and promoted.

Within this context, a consistent body of evidence has shown the beneficial effects of resistance training at improving physical and mental health aspects of older adults (El-Kotob et al., 2020; Fragala et al., 2019). Although a greater level of muscle strength is an important predictor of morbidity and mortality (Legrand et al., 2014), muscle power (i.e., ability to exert force rapidly) has emerged as a critical health indicator for older adults, especially because of its association with functional capacity (Foldvari et al., 2000; Bean et al., 2002). Not only this, but longitudinal studies have demonstrated that lower levels of muscle power are associated with an increased risk of hospitalization, all-cause mortality, and cognitive decline in these individuals (Losa-Reyna et al., 2022; Steves et al., 2016). As a consequence, improving muscle power in older adults has been highly advocated (Cadore & Izquierdo, 2018). As the muscle training effects are usually specific to the stimulus provided, it has been suggested that performing resistance exercises in an explosive manner, that is, with the concentric portion of the exercises performed as quickly as possible, would favor muscle power adaptation (Straight et al., 2016).

Known as high-velocity resistance training (HVRT), this mode of exercise has been shown to consistently increase functional capacity, as well as muscle strength (Englund et al., 2017; Schaun et al., 2021), and power (Byrne et al., 2016; Schaun et al., 2021) in older adults. Indeed, HVRT can lead to meaningful improvements in these outcomes while using only low to moderate training loads, which may facilitate the implementation of resistance training interventions to naïve exercisers and those with greater limitations, like mobility-limited older adults, who often consider themselves too unfit or unable to complete a standard exercise program. Such improvements in neuromuscular and physical function seem to be mediated by enhanced activation of the neural system during the exercise (Reid et al., 2015). The deterioration of neuromuscular activity with aging is strongly associated with a lower capacity to produce strength. Maximal handgrip strength, for example, is roughly 50% lower than expected in older adults, and this has been shown to be, amongst other factors, due to poor neuromuscular activation and motor unit recruitment (Kelly et al., 2018). Previous findings have indicated that enhancement in brain connectivity especially in regions such as the cortical and subcortical regions might be related to improvements in functional capacity, strength, and executive function (Chirles et al., 2017; Voss et al., 2010). Therefore, the favorable neuromuscular adaptations induced by HVRT might also be associated with improvements in cognitive function.

Nevertheless, compared to the extensive data on the effects of aerobic and traditional progressive resistance training in cognitive function (Christie et al., 2008; Dishman et al., 2006; Erickson I et al., 2019), little is known about the effects of HVRT on the executive function (Strassnig et al., 2015). Previous evidence suggest that exercise-induced muscle contraction may release myokines that are associated with improvements observed in cognition and increased neuroplasticity (Chow et al., 2022). More importantly, improvements in cognitive function have been associated with better quality of life, reduced risk of falls and mortality in those who are older (Brown & Landgraf, 2010; Johnson et al., 2007; Mirelman et al., 2012).

The domain of cognitive function involves several brain functions, including executive function. The latter represents a set of skills that underlie the basis of cognitive processes such as attention, inhibitory control, processing speed, and working memory (Bear et al., 2015). This set of skills are also evoked during the performance of multiple-task activities such as problem-solving (Bear et al., 2015). In older adults, decreased executive function is associated with reduced functional

capacity. This occurs mostly because these individuals' attention, problem-solving, and processing speed capacity become impaired, which prevents them from independently performing part of their daily life activities (Razani et al., 2007). In addition, impaired executive function significantly predicts the transition from mild cognitive impairment to dementia in older adults (Junquera et al., 2020). As such, improving or maintaining executive function may assist on retaining functional capacity and reduce the risk of incident dementia in adults.

However, to the best of our knowledge, no previous study has investigated the effect of HVRT on executive function in middle-aged and older adults with and without mobility limitations. These results can improve our understanding on how aging and functional capacity might be associated to the effects of exercise on executive function. Therefore, the aim of the present study was to examine the effect of HVRT on executive function in middle-aged and older adults with and without mobility limitations, and to test whether any improvements observed in functional capacity, strength and power would be associated with improvements observed on executive function. Our initial hypothesis was that 12 weeks of HVRT would improve the executive function of the participants, but with superior gains in the mobility-limited individuals. Also, we predicted that changes in executive function would be associated with changes in the muscular and functional capacity parameters, with muscular parameters mediating the association between functional capacity and executive function.

2. Methods

The present experimental, non-randomized study used secondary outcomes of a previous trial that aimed to investigate the effect of HVRT on the neuromuscular, morphological, and functional outcomes of middle-aged, healthy older, and mobility-limited older adults, which were previously published and made available elsewhere (Schaun et al., 2022). Herein we present the effect of HVRT on the cognitive outcomes of the abovementioned groups and test the association of these changes to those observed in select neuromuscular, morphological and functional outcomes, all of which are novel and not yet published.

2.1. Participants

Forty-five participants, including 19 middle-aged adults (40–55 years); 18 older adults (> 60 years) and 8 mobility-limited older adults (> 60 years) were included in the present study. A complete description of the participants' recruitment is available elsewhere (Schaun et al., 2022). Briefly, eligible and interested participants were invited for an in-person interview at the university, in which older participants performed the Short Physical Performance Battery (SPPB) battery and were objectively classified as being mobility-limited based on the SPPB score (≤ 9 points) (Guralnik et al., 1994). During the in-person interview individuals were additionally questioned about the exclusion criteria, which included cognitive impairment (Mini mental state examination < 23), diagnosis of progressive neurological disorders, musculoskeletal disorders, and osteoarthritis, stroke, revascularization, deep vein thrombosis or pulmonary embolism history in the past 12 months, severe cardiovascular disease, uncontrolled hypertension (> 160/90 mmHg), and current or previous participation in structured exercise more than 1x/week in the past six months. Informed consent was obtained from all participants and the research project was previously approved by the Federal University of Pelotas Ethics Review Board (CAAE: 13,459,719.0.0000.5313) in accordance with the Declaration of Helsinki. As previously explained, considering an $\alpha = 0.05$, 80% power, and an effect size of 0.22, a total sample of 54 participants (18 per group) was estimated (Schaun et al., 2022). In this study, we analyzed data from 45 participants with complete cognitive function evaluation at the baseline and post-intervention period.

2.2. Measurements

2.2.1. Outcomes

Executive function was assessed using four paper-pencil based tests performed before and after the 12-week intervention period. All assessments were carried out by trained researches who were blinded to the allocated group and study's main hypothesis. The tests were selected to encompass aspects related to visual search and processing speed, mental flexibility, working memory, cognitive interference, and verbal fluency. The scores from each test were converted into z-scores and adjusted for age, sex, education, diabetes, beta-blockers use, and total number of medications given their known associations with cognitive impairment.

2.2.1.1. Trial making test. The validated Portuguese version of the Trail Making Test (TMT) (Carvalho & Caramelli, 2020) was used to assess visual search, processing speed, and mental flexibility and includes two sections, the TMT-A and TMT-B. In the first, participants are requested to draw a line to connect 25 numbers in ascending order (from 1 to 25) and which are placed on a paper sheet in a non-organized manner. In the TMT-B, a similar task is performed, with the exception that instead of only numbers participants are requested to draw a line connecting numbers (from 1 to 3) and letters (from A-L) in ascending order in an alternating fashion. As an example, the initial order should be as follows: 1-A-2-B-3-C and so on. During both sections of the test the participants were asked to draw the lines as quickly as possible. Specifically, the TMT-B score is computed as the time in seconds that the participant takes to complete this specific task (a longer time indicate a worse cognitive performance), whereas the ratio between the time spent in the TMT-B and the TMT-A was used to assess mental flexibility (with lower values indicating better performance).

2.2.1.2. Digit span test. Working memory was assessed by the Digit Span Test (DST). During the test, participants are asked to verbally recall a series of numbers that are said by the researcher. The test begins with three numbers from 0 to 9, which are read with a 1 s interval between each number, and that the participant should recall in the correct order. In case the participant recalls them correctly, a new sequence is provided with an additional digit (i.e., four digits), and so on. The test ends when the participant is not able to recall the correct sequence two times in a row. The higher the number of digits recalled, the better the working memory (SCHROEDER et al., 2012). After the DST test, participants additionally performed a second task in which they were asked to recall the digits in a backward order, which is interpreted similar to the forward version.

2.2.1.3. Stroop color and word test. Cognitive interference was examined by the Portuguese version of the Stroop Color and Word test (Campanholo et al., 2014). Time to complete each task of the test was recorded to calculate the interference score based on the formula purposed by Golden et al. (1978). Lower scores indicated greater difficulty in inhibiting interference with a negative interference score revealing an inability to inhibit interference.

2.2.1.4. Language domain. Verbal fluency, another domain of executive function, was assessed using the FAS test (Williams et al., 2005). In this test, participants had 60 s to recall as many words as possible that began with the letters "F", "A", and "S", one letter at a time (60 s each). The total score was determined as the sum of the words recalled for each letter, in which higher scores are indicative of better verbal fluency.

Scores from all cognitive tests were standardized into z-scores based on data from the whole sample. An executive function domain score was created as the average of all standardized z-scores from each cognitive test, as previously recommended (Singh et al., 2014). Then, the averaged z-score was z-scored again. Higher scores indicated better

executive function.

2.2.2. Covariates

2.2.2.1. Maximal dynamic strength and peak power. Maximal dynamic strength was measured on the bilateral knee extension (KE) and leg press (LP) exercises (New Fitness®, São Paulo, Brazil) using the one-repetition maximum (1RM) test. Participants first performed a 10 repetitions warm-up set at a load equivalent to 50% of their estimated 1RM, which was determined during familiarization, and the resistance was increased until only one repetition could be performed using the appropriate technique and range of motion. After each attempt, resistance was adjusted based on the Lombardi scale (1989). 1RM load (in kg) was determined in no more than five attempts separated by 3 min intervals.

Twenty to thirty minutes after completion of the 1RM tests, participants performed one repetition of the KE exercise at 70% 1RM. For this repetition subjects were instructed to perform the concentric phase as fast possible and the eccentric phase in ~2 s and peak power was determined during the concentric portion of the exercises using a linear position transducer (Chronojump BoscoSystem®, Barcelona, Spain), which was attached to the weight stack on the KE machine.

2.2.2.2. Maximal isometric voluntary contraction. Participants were also tested for their maximal unilateral knee extension voluntary isometric contraction (MVIC) using a load cell (Miotec®, Porto Alegre, Brazil) with 200 kg capacity. Specifically, they were positioned on a knee extension machine with 90° of hip and knee flexion, and were instructed to exert maximal force as fast as possible while receiving strong verbal encouragement. Each participant had three attempts lasting ~5 s, and 2 min were allowed between attempts. The parameter used was maximal isometric force (kg), determined as the highest 1-s epoch of the force-time signal.

2.2.2.3. Muscle thickness. Five images from the rectus femoris (RF), vastus intermedius (VI), vastus lateralis (VL), and vastus medialis (VM) muscles of the right limb were then obtained by B-mode ultrasound (Tosbee/SSA-240a, Toshiba®, Japan) using a 7.5 MHz linear-array probe positioned perpendicularly to the muscle of interest. RF, VI, and VL images were obtained at 50% of the distance between the lateral condyle and the greater trochanter of the femur, whereas VM images were obtained at 20% of the same distance (Korhonen et al., 2009; Kumagai et al., 2000). Images were analyzed using the ImageJ software v1.52a (National Institutes of Health, Bethesda, USA). RF, VL, and VM muscle thicknesses were defined as the distance between the superficial and deep muscle aponeurosis, whereas VI thickness corresponded to the distance between the superficial aponeurosis and the hyperechoic interface of the femur. Muscle thickness was determined for each individual muscle as the average of all five images and summed to determine a measure of quadriceps muscle thickness.

2.2.2.4. Functional performance. Data from three functional tests were included in the present analysis. Lower limb strength was tested using the 30-s Sit to Stand (30-s STS) test. Participants were instructed to rise as many times as possible from a chair in 30 s and the total number of repetitions was used for the analysis. Agility and dynamic balance were assessed based on the Timed Up and Go (TUG) test, in which participants were instructed to get up from a chair, walk as fast as possible without running towards a cone positioned 3-m away, get around the cone and return to the initial position, sitting on the chair. The fastest time between two attempts was used in the analysis. Finally, participants performed a 10-m walking test with a numeric task (i.e., dual-task test). Specifically, they were requested to walk 10-m at their maximal gait speed, while simultaneously counting backwards from 100 by skipping 3 (e.g., 100, 97, 94, and so on). The corresponding gait speed was then determined in $m\ s^{-1}$.

2.3. Training program

After being tested for all outcomes of interest at baseline, participants completed a supervised 12-week HVRT intervention (June to November 2019) at a frequency of two sessions per week (for a detailed description of the training program see [Schaun et al., 2022](#)). Briefly, sessions included a 10-min warm-up containing general and specific warm-up activities with submaximal loads, followed by 1–3 sets of 8–10 repetitions of 5 different exercises: LP, KE, seated plantar flexion, horizontal chest press and seated row. Each week, the first and second sessions were performed at 40% and 60% 1RM, respectively, and participants were instructed to perform the concentric phase of each repetition as fast as possible and the eccentric phase in approximately 2 s. Two minutes of rest were allowed between each set and exercise. To account for possible adaptations in strength, exercise loads were increased by 5% after the sixth week of training. Sessions took place at the fitness center of the Federal University of Pelotas, Brazil.

2.4. Statistical analysis

Continuous variables were reported as mean $\pm$ SD, whereas categorical outcomes were reported in absolute and relative terms. Baseline differences were compared using the Fisher's exact test, one-way ANOVA with Bonferroni's post hoc, and Kruskal-Wallis test followed by Dunn's post hoc, as specified.

All statistical models were adjusted for age and sex, as well as education, use of beta-blockers, diagnosed diabetes, and number of prescribed medications for models including cognitive outcomes.

Training-related adaptations in cognitive measures were then compared between groups (middle-aged, older adults, and mobility-limited older adults), time points (pre- and post-training), and group*time point interactions using a Generalized Estimating Equation (GEE)

model. Intention-to-treat analysis was performed including all participants from pre-training in the model. Difference between time points were reported as adjusted marginal mean differences (AMMD).

Changes in functional capacity and muscular parameters were previously reported ([Schaun et al., 2022](#)) and are included in the Suppl. File (Table S1). Change values were then fitted into a multiple linear regression model to test the linear associations between changes in executive function outcomes and those in functional capacity and neuromuscular fitness parameters. In this analysis, the baseline score of the dependent variable was included in the model.

Next, a mediation analysis was used to estimate the indirect effects of changes in chair stand and dual-task test performance (mediated through changes in lower body muscular parameters) on changes in cognitive function while accounting for the direct effect functional performance has on increases in lower-body muscular parameters and cognitive function. These variables were selected because of the significant associations between lower body strength and these dependent variables in multiple linear regression models ([Figs. 1 and 2](#)). This method provides unstandardized estimates of the direct and indirect relationships, along with their corresponding 95% confidence intervals (CI). Mediation analyses included models with changes in executive domain as the dependent variables and changes in lower body strength (LP, KE, and MVIC), power (peak power at 70% 1RM), and quadriceps muscle thickness as the mediating variables. To determine reverse causality, a second model included change in lower-body muscular parameters as the dependent variable and change in executive function as the mediating variable. Results from mediation analysis were considered significant if the CIs did not include 0. All regressions were performed using whole-sample data. Results were reported as unstandardized estimates of the direct and indirect pathways, along with corresponding 95% CIs. Finally, statistical analyses were performed using STATA/MP 14.2 (StataCorp, College Station, Texas) with an alpha equal to 5%.

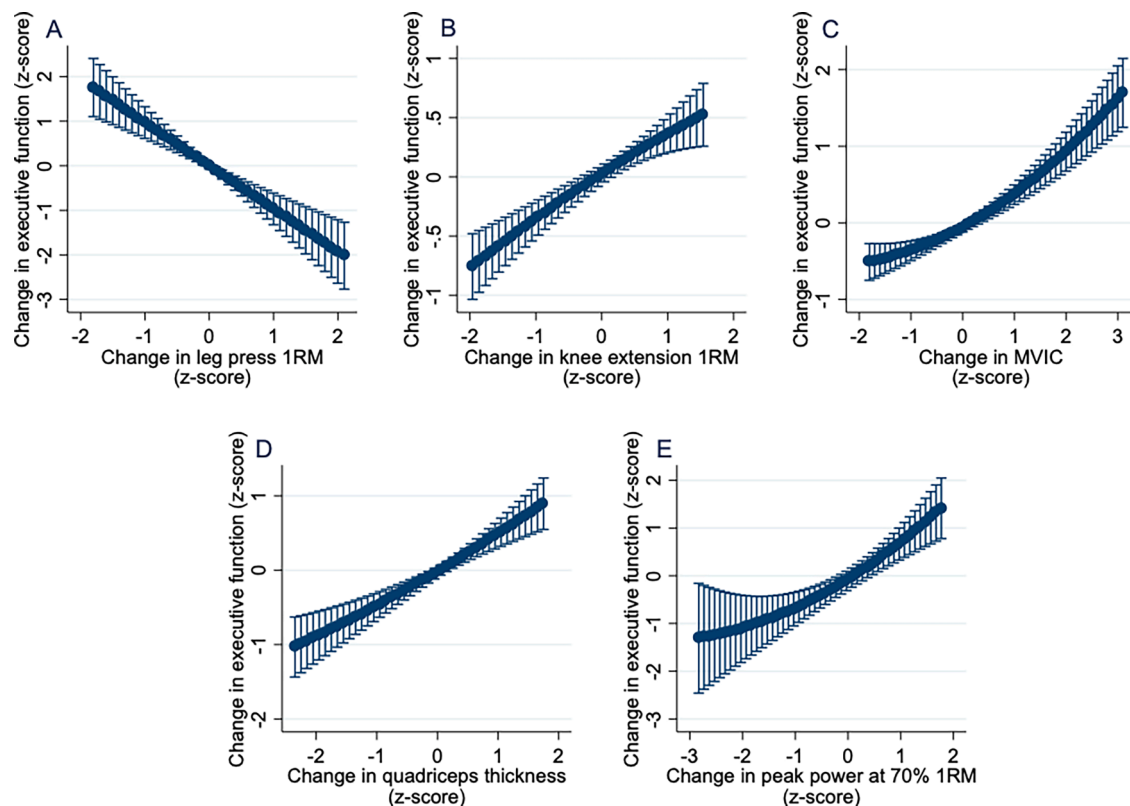


Fig. 1. Associations between changes in lower body strength, power, and muscular thickness and changes in executive function. Values are expressed in z-scores and 95% confidence interval and adjusted for age, educational level, beta-blocker use, diagnosed diabetes, number of medications, and baseline executive function. 1RM: one repetition maximum; MVIC: maximal voluntary isometric contraction.

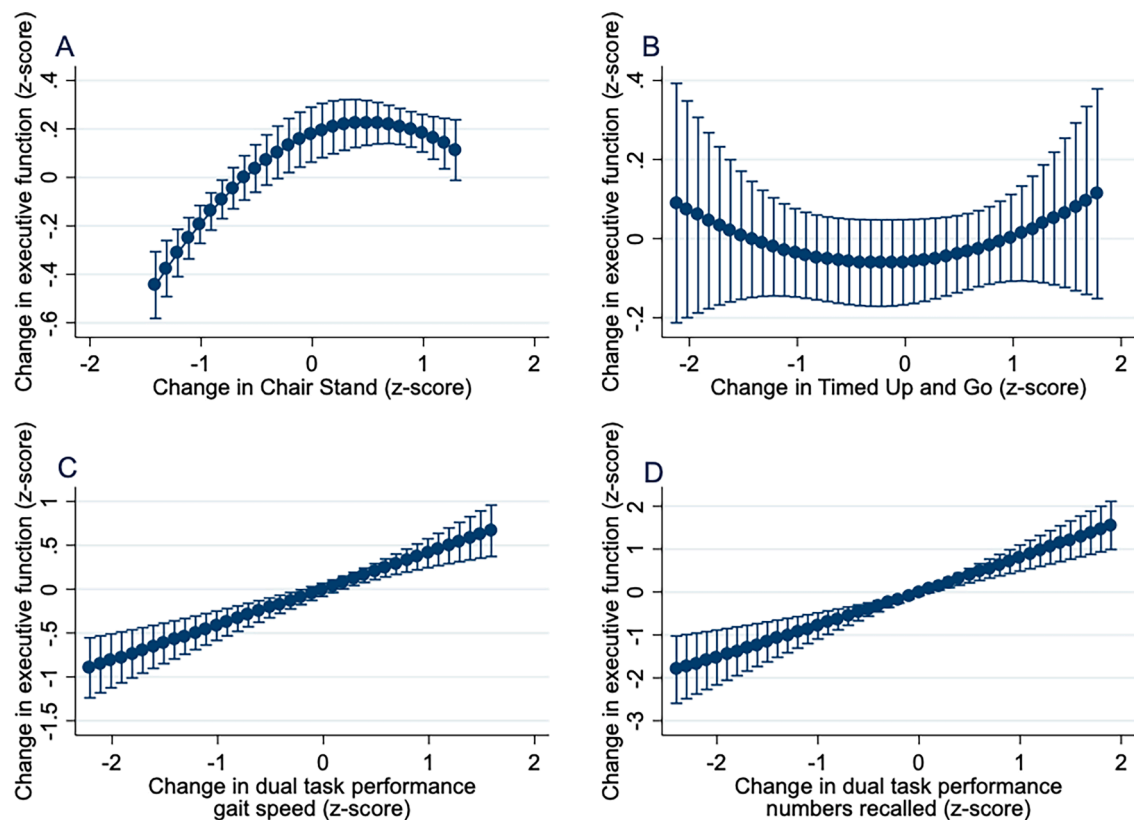


Fig. 2. Association between changes in functional capacity and changes in executive function. Values are expressed in z-scores and 95% confidence interval and adjusted for age, educational level, beta-blocker use, diagnosed diabetes, number of medications, and baseline executive function.

3. Results

3.1. Baseline

Out of the 45 participants (female: $n = 22$, 48.9%), who were assessed at baseline for the cognitive outcomes, 41 completed the 12-week exercise training intervention and follow-up assessments, as shown in the Suppl. File (Figure S1). As shown in Table 1, the middle-

Table 1
Characteristics of the participants at baseline. $N = 45$.

	Middle age ($n = 19$)	Older adults ($n = 18$)	Mobility-limited older adults ($n = 8$)	P value
Mean age, year (SD)	48.5 (5.02) ^{b,c}	68.0 (6.07) ^{a,c}	77.3 (7.39) ^{a,b}	<0.001 ¹
Sex, n (%)				0.628 ²
Male	11 (57.9)	9 (50.0)	3 (37.5)	
Female	8 (42.1)	9 (50.0)	5 (62.5)	
Education, years (SD)	13.9 (3.20) ^c	12.5 (3.33)	8.8 (5.33)	0.009 ¹
Ethnicity, n (%)				0.647 ²
White	16 (84.2)	18 (100.0)	8 (100.0)	
Non-white	3 (15.8)	0 (0.0)	0 (0.0)	
Marital status, n (%)				0.248 ²
Single or divorced	7 (36.8)	6 (33.3)	2 (25.0)	
Married	12 (63.2)	11 (61.1)	4 (50.0)	
Widowed	0 (0.0)	1 (5.6)	2 (25.0)	

^a different from the middle-aged adults;.

^b different from the older adults;.

^c different from the older adults with mobility limitation.

¹ One-way ANOVA with Bonferroni's post hoc.

² Fisher's exact test.

aged group had more male and non-white participants compared to the other groups at baseline, whereas the mobility-limited older adults' group participants were older than the other two groups, as presented in Table 1.

3.2. Training responses

As can be observed in Table 2, HVRT improved TMT-B performance (AMMD: -38.60; 95%CI: -66.22; -10.98; $p = 0.006$), as well as Executive Function in the mobility-limited older adults' group (AMMD: 0.21; 95% CI: 0.04, 0.38; $p = 0.040$). However, this latter difference was not evident neither in the middle-aged (AMMD: 0.03; 95%CI: -0.09; 0.17; $p = 0.533$) nor in the older adults (AMMD: -0.11; 95%CI: -0.25; 0.02; $p = 0.107$) groups. Furthermore, TMT-A performance improved in the three groups whereas TMT B/A ratio, incongruent score, digit span test, and verbal fluency was not altered by the intervention in any of the groups.

3.3. Multiple linear regression analyses

The associations between training-related changes in executive function with both neuromuscular and functional capacity parameters are shown in Figs. 1 and 2, respectively. Whole-sample regression analysis revealed that improvements in LP (β : -0.93; 95%CI: -1.13; -0.73) and KE 1RM (β : 0.37; 95%CI: 0.17; 0.56), as well as in MVIC (β : 0.36; 95%CI: 0.25; 0.49), quadriceps muscle thickness (β : 0.48; 95%CI: 0.31; 0.65), and peak power at 70%1RM (β : 0.69; 95%CI: 0.39; 0.99) were all associated with changes in executive function (Fig. 1). As for the functional performance measures (Fig. 2), changes in chair stand test performance (β : 0.18; 95%CI: 0.12; 0.25), gait speed (β : 0.41; 95%CI: 0.26; 0.57), as well as in the total numbers recalled during the 10-m gait speed test (β : 0.79; 95%CI: 0.49; 1.09) were also associated with changes in executive function, whereas changes in the TUG test performance were not (β : 0.02; 95%CI: -0.06; 0.10).

Table 2High-velocity resistance training effects (mean $\pm$ SD) on cognitive function in middle-age and older adults and mobility-limited older adults. $N = 45$.

	Middle age		Older adults		Mobility-limited older adults		Group p	Time p	Group*Time p
	Pre	Post	Pre	Post	Pre	Post			
Cognitive function									
Trail Making test, sec									
TMT-A, seconds	30.45 (8.05)	26.22 [†] (9.13)	53.01 (17.69)	45.49 [†] (17.57)	81.57 (46.22)	69.77 [†] (38.49)	0.856	0.002	0.542
TMT-B, seconds	74.64 (50.64)	62.85 (43.04)	119.24 (62.65)	127.45 (80.81)	227.72 (105.43)	189.12 [†] (100.02)	0.306	0.022	0.047
TMT-B/TMT-A, ratio	2.38 (1.45)	2.31 (1.00)	2.33 (1.12)	2.79 (1.35)	3.06 (1.49)	2.74 (0.86)	0.771	0.492	0.445
Incongruent score, ratio	48.97 (22.27)	55.95 (10.75)	49.31 (29.71)	41.77 (23.11)	25.36 (24.14)	27.91 (22.20)	0.748	0.798	0.157
Digit Span Test, digits	9.53 (2.20)	9.29 (3.10)	7.94 (2.73)	8.06 (2.51)	6.63 (2.61)	6.87 (2.53)	0.361	0.699	0.965
Verbal fluency, words	39.95 (11.94)	42.65 (13.87)	34.67 (13.59)	36.13 (14.96)	26.25 (11.55)	29.87 (18.22)	0.742	0.165	0.724
Executive function, z-score	0.23 (0.32)	0.26 (0.39)	0.01 (0.46)	-0.09 (0.45)	-0.58 (0.67)	-0.37 [†] (0.68)	<0.001	0.306	0.005

Values were adjusted for age, sex, and educational level. Bold values indicate statistical significance ($p < 0.05$).

SD: standard deviation. TMT: trail making test; sec: seconds;

[†] Significant difference from baseline (adjusted p-value < 0.05).

3.4. Mediation analyses

Because changes in both neuromuscular and functional performance

Table 3Mediation analyses of the effect of changes in functional performance on cognitive function through changes in lower body muscle strength, power, and muscle thickness. $N = 41$.

	Direct effect		Indirect effect
<i>Δ Executive Function</i>			
Δ Chair stand test			
	0.80 (0.66; 0.94)	Δ Leg press	-0.27 (-0.41; -0.13)
	0.90 (0.72; 1.08)	Δ knee extension	-0.38 (-0.57; -0.18)
	0.50 (0.35; 0.64)	Δ MVIC	0.03 (-0.02; 0.08)
	0.66 (0.51; 0.82)	Δ Quadriceps thickness	-0.13 (-0.25; -0.02)
	0.57 (0.43; 0.70)	Δ Power	-0.04 (-0.09; 0.01)
Δ Dual-Task test - gait speed			
	0.21 (-0.05; 0.47)	Δ Leg press	-0.07 (-0.23; 0.10)
	-0.59 (-1.51; 0.33)	Δ knee extension	0.73 (-0.18; 1.65)
	0.04 (-0.18; 0.26)	Δ MVIC	0.11 (-0.00; 0.21)
	0.72 (-0.21; 1.64)	Δ Quadriceps thickness	-0.57 (-1.50; 0.35)
	0.14 (-0.06; 0.35)	Δ Power	0.00 (-0.02; 0.03)
Δ Dual-Task test-recalled numbers			
	0.26 (0.00; 0.52)	Δ Leg press	-0.09 (-0.27; 0.08)
	0.15 (-0.05; 0.35)	Δ knee extension	0.02 (-0.02; 0.06)
	0.87 (0.67; 1.07)	Δ MVIC	-0.70 (-0.92; -0.47)
	0.22 (0.02; 0.43)	Δ Quadriceps thickness	-0.05 (-0.13; 0.02)
	2.15 (1.77; 2.53)	Δ Power	-1.98 (-2.49; -1.47)

Bold values indicate statistical significance ($p < 0.05$).

Values are expressed in unstandardized estimates with 95% confidence intervals.

Δ: changes from baseline.

MVIC: maximum voluntary isometric contraction.

parameters were associated with changes in executive function, mediation analyses were performed and are shown in Table 3. Specifically, changes in chair stand were found to exert an indirect effect on executive function through changes in maximal dynamic strength (i.e., LP and KE 1RM) and quadriceps muscle thickness. Likewise, changes in MVIC and peak power seemed to mediate the indirect effect that changes in the amount of numbers recalled during the dual task test had on executive function. No direct and indirect effect of gait speed on executive function was observed. A significant indirect effect of changes in chair stand performance on LP 1RM (indirect effect: β : 0.21; 95%CI: 0.05, 0.38) and knee extension 1RM (indirect effect: β : -0.16; 95%CI: -0.29, -0.03) through changes in executive function indicates possible reverse causality in this association. However, changes in executive function did not mediate the effect of chair stand performance on quadriceps muscle thickness (indirect effect: β : -0.13; 95%CI: -0.29, 0.03, direct effect: β : 0.67; 95%CI: 0.39, 0.95), MVIC (indirect effect: β : -0.16; 95%CI: -0.29, -0.03, direct effect: β : 0.07; 95%CI: -0.05, 0.18), and peak power (indirect effect: β : -0.03; 95%CI: -0.09, 0.02, direct effect: β : 0.99; 95%CI: 0.92, 1.05), reducing the likelihood of a reverse causality effect.

4. Discussion

Overall, our findings suggest that a short-term HVRT intervention was capable of improving the executive function of older adults with mobility limitations. Our results further demonstrate that improvements in executive function were associated to neuromuscular and the functional performance parameters, while the mediation analyses suggest that the improvements in executive function through functional performance were mediated by improvements in the neuromuscular parameters.

As previously mentioned, executive function encompasses core cognitive processes including inhibitory control, working memory, attention, and cognitive flexibility (Miyake et al., 2000). Thus, impairments in this cognitive domain may negatively affect functional capacity (Vaughan & Giovanello, 2010) and increase the risk of falling (Kearney et al., 2013) and developing dementia (Jung et al., 2020) in older individuals. Here, we showed possibly for the first time that a HVRT program improved executive function in mobility-limited older adults, albeit no effect was observed in middle-aged and older participants with preserved functional capacity.

Traditional progressive resistance training has been suggested as a key factor to improve muscle mass, strength, mobility, cognition, as well as the functional capacity of older adults (Izquierdo et al., 2021; Schaun et al., 2021). Indeed, prior systematic reviews and meta-analyses have

shown that resistance training improved functional capacity (Liu & Latham, 2009) and executive function (Gates et al., 2013) in healthy older adults and in older adults with mild cognitive impairment, respectively. In the latter, the authors suggested that high-load resistance training might result in greater improvements in executive function as compared to low-load resistance training or aerobic training. A previous randomized controlled trial showed that six months of high-load ($\geq 80\%$ 1RM) progressive resistance training increased the executive function (mean z-score change: 0.42; 95%CI: 0.25–0.60) of older adults with mild cognitive impairment (Singh et al., 2014). Although greater than the effect observed in the mobility-limited older adults of the present study (mean z-score change: 0.21; 95%CI: 0.04–0.38) (Table 2), both effect sizes are greater than that observed for low-to-moderate aerobic training (mean z-scores: 0.12; 95%CI: 0.02–0.23) (Smith et al., 2010). Given that our sample had a relatively preserved cognitive function (i.e., all participants had a MMSE score ≥ 24), it is expected that older adults with mild cognitive impairment would present a greater improvement in cognitive function in response to training.

Maybe more importantly, HVRT has been consistently shown to improve muscle power and functional capacity (Schaun et al., 2021) while using only low to moderate training loads (~ 40 – 60% 1RM). This is considerably lower than what is typically advocated for traditional progressive resistance training (i.e., $\geq 80\%$ 1RM; Fragala et al., 2019). Such difference may facilitate the implementation of resistance training interventions to naïve exercisers and those with greater limitations, like mobility-limited older adults. Based on our results, this can be true not only in order to improve physical parameters, but also to assist on the maintenance of cognitive function, at least in mobility-limited older individuals.

A plausible explanation of the improvements induced by HVRT on executive function may be related to the high movement velocities observed in this exercise mode. Muscle contraction releases neurogenic myokines including irisin and cathepsin B (Pedersen, 2019). These substances stimulate the expression of BDNF in the hippocampus that are responsible for inducing improvements on the brain's structural and functional plasticity that benefit cognitive function (Pedersen, 2019). Moreover, the myokine response magnitude is believed to be related to the volume of muscle mass recruited, and previous data suggest that explosive muscle contractions such as those performed during HVRT may elicit a similar or even greater recruitment of type II fibers than that observed during traditional high-load resistance training (Englund et al., 2017). Indeed, in a sample of older women with preserved functional capacity and without dementia, improvements in cognitive function were similar between traditional resistance training performed at moderate-to-high loads and power training performed at low-to-moderate loads (Coelho-Júnior et al., 2020). Future robust randomized controlled trials are warranted to confirm or refute this hypothesis.

Although mobility-limited older participants improved their executive function after HVRT, no significant effect was observed for the middle-aged and healthy older adults' groups in the present investigation (Table 2). Considering that these participants had a relatively preserved cognitive function (i.e., all participants had a MMSE score ≥ 24), were not functionally impaired, and that there is a well-known mediation effect of muscle strength on the cognitive benefits observed in response to resistance training (Mavros et al., 2017), our hypothesis is that this may be due to a plateau in the capacity of functional performance improvements to improve executive function. As observed in Fig. 2A, changes in executive function were linearly associated with changes in chair stand test performance until a certain threshold (a z-score of ≈ 0), where such association was apparently absent. Although speculative, it is possible that after a certain level, other mechanisms including contraction-dependent myokines, are linked to the maintenance of executive function in such individuals. In a previous randomized controlled trial, lower-body strength mediated the effect training on

global cognition, but not on executive function (Mavros et al., 2017). These results led the authors to propose that muscle contraction is linked to cognitive improvement. Indeed, irisin, a myokine identified as a potential target for Alzheimer's diseases treatment (Lourenco et al., 2019), was shown to be associated with skeletal muscle hypertrophy (Reza et al., 2017).

Furthermore, we observed that changes in LP 1RM were inversely correlated with changes in executive function (Fig. 1A). Although this finding was not expected, some possible explanations are required. For example, changes in the KE 1RM ($r = -0.77$; $p < 0.001$) and quadriceps thickness ($r = -0.50$; $p = 0.001$) were also inversely associated with changes in leg press 1RM in the present study. Participants with lower quadriceps thickness and KE strength may also improve other muscles involved in the performance of the LP exercise (e.g., gluteus), which in turn would reduce the specific effect of this exercise on the knee extensors.

Changes in the gait speed and on the amount of numbers recalled during the dual-task test in response to training were also associated with changes in executive function (Fig. 2C and 2D). Gait speed is a well-recognized marker of general health and functional status, especially in older individuals (Van Kan et al., 2009), being strong and independently associated with mortality risk (Studenski et al., 2011) and cognitive function (Peel et al., 2019). Although no neuromuscular factor was identified as a potential mediator of the effect of gait speed itself on cognitive performance, muscle power and MVIC mediated the association between the change in the amount of numbers recalled during the dual-task test and changes in executive function. The ability to perform multiple tasks while engaged in a concurrent mental task (e.g., walking and talking) clearly becomes impaired with aging (Al-Yahya et al., 2011). A key factor that underlies this impairment is the disproportionate age-related decrease in executive function (Reuter-Lorenz et al., 2021), which corroborates the observed association between the change in the amount of numbers recalled during the test and the changes observed in executive function. This association was also partially mediated by changes in the MVIC and muscle power. Previous studies have demonstrated that gains in lower-limb muscle strength are associated with improvements in the performance of dual-task activities, as well as in the performance of other functional tests including the TUG (Casas-Herrero et al., 2013; Zak et al., 2021), corroborating our findings. Reid et al. suggested that lower-body muscle strength was an important determinant of functional capacity in mobility-limited older adults (Reid et al., 2008). Likewise, Casas-Herrero et al. (2013) reported a significant correlation between functional capacity, muscle mass, muscle strength, and muscle power in frail individuals, and in those with combined frailty and mild cognitive impairment.

Some limitations from the present study must be acknowledged. First, caution is needed when extrapolating our results to severely mobility-limited individuals, as our sample included only individuals with mild to moderate limitations in mobility. Second, although we were unable to reach the estimated sample size of 18 participants in the mobility-limited older adults' group, the statistical model adopted (i.e., GEE) appeared capable of differentiating this group from the others. It was also able to highlight the presence (or lack thereof) of training-related effects, when they differed from the other groups. Also, the recruitment phase and in-person activities were suspended due to the COVID-19 pandemic, so it was not possible to finish recruiting of the mobility-limited participants and reach the required sample size. Finally, the exclusion of participants with a MMSE score < 23 should be kept in mind as this may have affected the reported associations in the training-related responses.

5. Conclusion

A short-term HVRT intervention improved the executive function of mobility-limited older adults. Such improvements seemed mediated by changes in lower-body muscle strength, power, and quadriceps muscle

thickness. Future studies are warranted to examine the biological pathways linking these muscle parameters and cognitive function.

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Author agreement

All authors have approved the final manuscript.

Statement of human and animal rights

This study was approved by the Federal University of Pelotas Ethics Review Board (CAAE: 13459719.0.0000.5313) in accordance with the Declaration of Helsinki.

Consent to participate

Informed consent was obtained from all subjects involved in the study.

CRediT authorship contribution statement

Natan Feter: Conceptualization, Investigation, Formal analysis, Methodology, Writing – original draft. **Gustavo Z. Schaun:** Conceptualization, Investigation, Methodology, Writing – review & editing. **Emily C. Smith:** Writing – review & editing. **Júlia Cassuriaga:** Methodology, Writing – review & editing. **Ricardo Alt:** Methodology, Writing – review & editing. **Larissa Redig:** Methodology, Writing – review & editing. **Cristine Lima Alberton:** Conceptualization, Methodology, Funding acquisition, Writing – review & editing. **Jeff S. Coombes:** Writing – review & editing. **Airton J. Rombaldi:** Writing – review & editing.

Declaration of Competing Interest

The authors report no biomedical financial interests or potential conflicts of interest.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.archger.2023.105081](https://doi.org/10.1016/j.archger.2023.105081).

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